$= f(\sum_{i=1}^{m} \alpha_i x_i + \alpha_{m+1} x_{m+1})$

$= f((1-\alpha_{m+1}) x_0 + \alpha_{m+1} x_{m+1})$, $x_0 = \sum_{i=1}^{m} \frac{\alpha_i}{1-\alpha_{m+1}} x_i$

④ $\because \sum_{i=1}^{m} \frac{\alpha_i}{1-\alpha_{m+1}} = 1$, $\frac{\alpha_i}{1-\alpha_{m+1}} \in (0,1)$, S convex

$\therefore S$ contains x_0 (by 2)

$\therefore f(x_0) = f(\sum_{i=1}^{m} \frac{\alpha_i}{1-\alpha_{m+1}} x_i) \leq \sum_{i=1}^{m} \frac{\alpha_i}{1-\alpha_{m+1}} f(x_i)$ (by ②)

⑤ $\because f(\sum_{i=1}^{m+1} \alpha_i x_i) = f((1-\alpha_{m+1}) x_0 + \alpha_{m+1} x_{m+1})$

$\leq (1-\alpha_{m+1}) f(x_0) + \alpha_{m+1} f(x_{m+1})$ (by ①)

$\leq (1-\alpha_{m+1}) (\sum_{i=1}^{m} \frac{\alpha_i}{1-\alpha_{m+1}} f(x_i)) + \alpha_{m+1} f(x_{m+1})$

$= \sum_{i=1}^{m+1} \alpha_i f(x_i)$ ▨

6. strict discrete Jensen's inequality

6.2 real-valued $f(x)$ strictly convex on S, S convex set $\subseteq V$, V real vector space

$\Leftrightarrow f(\sum_{i=1}^{n} \alpha_i x_i) < \sum_{i=1}^{n} \alpha_i f(x_i)$, $\forall n \geq 2$,

$\forall x_i \in S$, $i = 1, \cdots, n$, $\sum_{i=1}^{n} \alpha_i = 1$, $\alpha_i \in (0,1)$.

$\exists k, j$, $x_k \neq x_j$

proof: $\Leftarrow$: let $n=2 \Rightarrow f$ strictly convex

$\Rightarrow$: ①②③ (same as that of 5.3, but use $<$) ④ if $\exists k, j \leq m$, $x_k \neq x_j \Rightarrow f(x_0) < \sum_{i=1}^{m} \frac{\alpha_i}{1-\alpha_{m+1}} f(x_i)$ (by ②)

else: $x_1 = x_2 = \cdots = x_m \Rightarrow x_0 \neq x_{m+1}$

$\Rightarrow f((1-\alpha_{m+1}) x_0 + \alpha_{m+1} x_{m+1})$ ③ then, $f(\sum_{i=1}^{m+1} \alpha_i x_i)$

⑤ $\therefore$ one of the two $\leq$ in 5.3, ⑤ is $<$. ▨

3. real-valued function $f(x)$ convex on convex set $S \subseteq V$, V real vector space,

$\Leftrightarrow f(\alpha x + (1-\alpha) y) \leq \alpha f(x) + (1-\alpha) f(y)$

$\forall x, y \in S$, $\forall \alpha \in (0,1)$

4 real-valued $f(x)$ strictly convex on S, S convex set $\subseteq V$, V real vector space.

$\Leftrightarrow f(\alpha x + (1-\alpha) y) < \alpha f(x) + (1-\alpha) f(y)$,

$\forall x, y \in S$, $x \neq y$, $\forall \alpha \in (0,1)$

5.1 real-valued function $f(x)$ convex on S, S convex set $\subseteq V$, V real vector space,

$\Leftrightarrow f(\sum_{i=1}^{n} \alpha_i x_i) \leq \sum_{i=1}^{n} \alpha_i f(x_i)$, $\forall n \geq 2$,

$\forall x_i \in S$, $i = 1, 2, \cdots, n$.

$\sum_{i=1}^{n} \alpha_i = 1$, $\alpha_i \in (0,1)$.

5.3 proof: $\Leftarrow$: let $n=2, \Rightarrow f$ convex

$\Rightarrow$: ① f convex $\Rightarrow f(\sum_{i=1}^{2} \alpha_i x_i) < \sum_{i=1}^{2} \alpha_i f(x_i)$ ② if $f(\sum_{i=1}^{m} \alpha_i x_i) \leq \sum_{i=1}^{m} \alpha_i f(x_i)$, $\forall x_i \in S$, $\sum_{i=1}^{m} \alpha_i = 1$, $\alpha_i \in (0,1)$

③ then, $f(\sum_{i=1}^{m+1} \alpha_i x_i)$

⑤ $\because \sum_{i=1}^{m+1} \alpha_i x_i$

$= (1-\alpha_{m+1}) x_0 + \alpha_{m+1} x_{m+1}$

$\in S$ (by ①) ▨

discrete Jensen's inequality

set S convex $\subseteq V$, V real vector space

$\Leftrightarrow \alpha x + (1-\alpha) y \in S$, $\forall x, y \in S$, $\forall \alpha \in (0,1)$

(Δ means by definition)

2. set S convex $\subseteq V$, V real vector space.

$\Leftrightarrow \sum_{i=1}^{n} \alpha_i x_i \in S$, $\forall n \geq 2$, $\forall x_i \in S$, $i = 1, 2, \cdots, n$.

$\sum_{i=1}^{n} \alpha_i = 1$, $\alpha_i \in (0,1)$

proof: $\Leftarrow$: let $n=2$, $\because S$ convex.

$\Rightarrow$: ① $n=2$, $\because S$ convex, $\therefore \sum_{i=1}^{2} \alpha_i x_i \in S$

② if $n=m$, $\sum_{i=1}^{m} \alpha_i x_i \in S$, $\sum_{i=1}^{m} \alpha_i = 1$, $\alpha_i \in (0,1)$

③ then, for $n = m+1$,

$\sum_{i=1}^{m+1} \alpha_i x_i = \sum_{i=1}^{m} \alpha_i x_i + \alpha_{m+1} x_{m+1}$

$= (1-\alpha_{m+1}) \sum_{i=1}^{m} \frac{\alpha_i}{1-\alpha_{m+1}} x_i + \alpha_{m+1} x_{m+1}$

$= (1-\alpha_{m+1}) x_0 + \alpha_{m+1} x_{m+1}$

④ $\because \sum_{i=1}^{m} \frac{\alpha_i}{1-\alpha_{m+1}} = 1$, $\frac{\alpha_i}{1-\alpha_{m+1}} \in (0,1)$, $\therefore x_0 = \sum_{i=1}^{m} \frac{\alpha_i}{1-\alpha_{m+1}} x_i \in S$ (by ②)